

LAB 3

Architecture Selection & Written Justification

Campus Placement & Internship Pipeline

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Architecture Selection: We chose Layered Architecture for the Campus Placement & Internship Pipeline system.

Architectural Choice: The system is organized into Presentation, Business, Security & Support, and Data layers. The Presentation layer provides the user interface, the Business layer handles placement drives, applications, resume and eligibility processing, interviews, offers, and notifications, while the Security & Support layer handles authentication and protection and the Data layer stores placement information.

Reason 1 – Clear separation of responsibilities: The placement system contains several distinct functions such as drive management, student applications, eligibility checking, interview management, and offer management. Separating these functions into layers makes the system easier to understand, maintain, and modify without affecting unrelated parts.

Reason 2 – Suitable for the current system structure: The Campus Placement & Internship Pipeline mainly follows a structured workflow from user interaction to business processing and database storage. A layered structure matches this flow and provides a clear path between the user interface, business components, security services, and stored placement data.

Security Advantage: A separate Authentication & Security Service provides a controlled point for authentication and security-related operations. This supports the requirement to protect sensitive academic records and offer documents using AES-256 encryption.

Performance Benefit: Separating business processing from presentation and data storage helps keep responsibilities focused and reduces unnecessary processing in the user interface. This supports efficient processing of eligibility requests and helps the system meet the requirement to display eligibility results within 3 seconds for at least 95% of requests under normal conditions.

Conclusion: Layered Architecture is therefore appropriate for the Campus Placement & Internship Pipeline because it provides clear organization of system responsibilities, supports security controls, and provides a structured path for efficient processing.